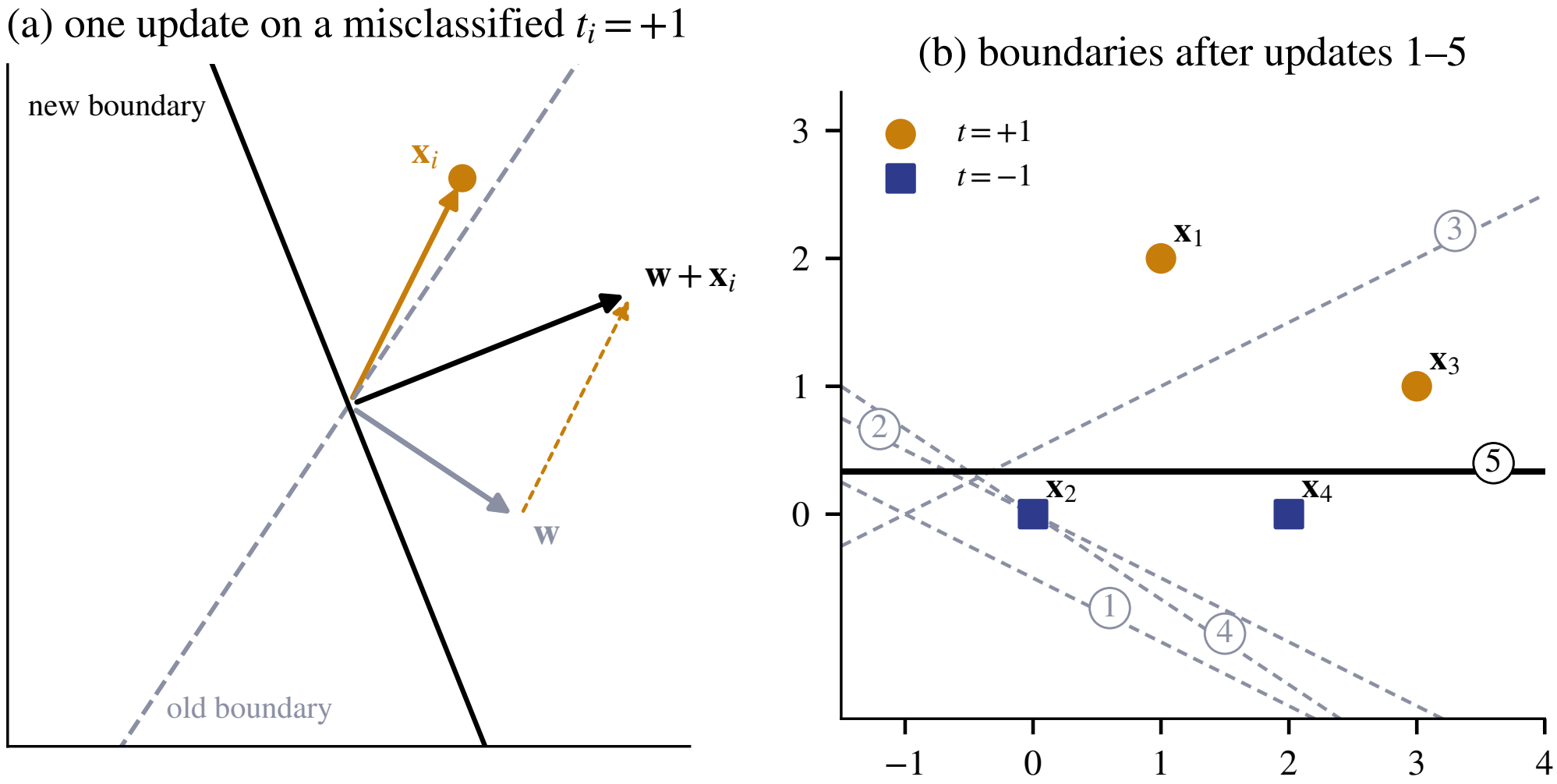
(a) one update on a misclassified $t_i = +1$
new boundary
$\mathbf{x}_i$
$\mathbf{w} + \mathbf{x}_i$
$\mathbf{w}$
old boundary
(b) boundaries after updates 1–5
$t = +1$
$t = -1$
$\mathbf{x}_1$
$\mathbf{x}_2$
$\mathbf{x}_3$
$\mathbf{x}_4$
1
2
3
4
5
3
2
1
0
−1
0
1
2
3
4